IEOR 160 : Homework 02

1.) 2nd order approx of

$$f(x) = e^{x_1 - 2x_2} + \cos(x_1 + 5x_2) - 2\sin(x_1)\cos(x_2) \quad \text{around } (0,0)$$

$$\frac{\partial}{\partial x_1} f(x) = e^{x_1 - 2x_2} - \sin(x_1 + 5x_2) - 2\cos(x_1)\cos(x_2)$$

$$\frac{\partial}{\partial x_2} f(x) = -2e^{x_1 - 2x_2} - 5\sin(x_1 + 5x_2) + 2\sin(x_1)\sin(x_2)$$

$$\nabla f(x) = \begin{bmatrix} e^{x_1 - 2x_2} - \sin(x_1 + 5x_2) - 2\cos(x_1)\cos(x_2) \\ -2e^{x_1 - 2x_2} - 5\sin(x_1 + 5x_2) + 2\sin(x_1)\sin(x_2) \end{bmatrix}$$

$$\nabla^2 f(x) = \begin{bmatrix} e^{x_1 - 2x_2} - \cos(x_1 + 5x_2) + 2\sin(x_1)\cos(x_2) & -2e^{x_1 - 2x_2} - 5\cos(x_1 + 5x_2) + 2\cos(x_1)\sin(x_2) \\ -2e^{x_1 - 2x_2} - 5\cos(x_1 + 5x_2) + 2\cos(x_1)\sin(x_2) & 4e^{x_1 - 2x_2} - 25\cos(x_1 + 5x_2) + 2\sin(x_1)\cos(x_2 \end{bmatrix}$$

$$f(x) \approx f(0) + \nabla f(0)\, x + \frac{1}{2} x^T \nabla^2 f(0)\, x$$

$$= 2 + \begin{bmatrix} -1 \\ -2 \end{bmatrix} x + \frac{1}{2} x^T \begin{bmatrix} 0 & -7 \\ -7 & -21 \end{bmatrix} x$$

$$= 2 - 1x_1 - 2x_2 - 7x_1x_2 - \frac{21}{2}x_2^2$$

2.) $f(x_1, x_2) = x_1^4 - x_1 x_2 + x_2^4$

$$\nabla f(x_1, x_2) = \begin{bmatrix} 4x_1^3 - x_2 \\ 4x_2^3 - x_1 \end{bmatrix}^T$$

foc

$$\nabla f(x_1, x_2) = 0$$

$$4x_1^3 - x_2 = 0$$

$$4x_2^3 - x_1 = 0$$

$$x_1 = 4x_2^3$$

$$4(4x_2^3)^3 - x_2 = 0$$

$$4(64x_2^9) - x_2 = 0$$

$$256x_2^9 - x_2 = 0$$

$$x_2(256x_2^8 - 1) = 0$$

$$x_2 = 0, \quad x_2 = \pm\frac{1}{2}$$

$$x_1 = 0, \quad x_1 = \pm\frac{1}{2}$$

$$x = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$$

SoC

$$\nabla^2 f(x) = \begin{bmatrix} 12x_1^2 & -1 \\ -1 & 12x_2^2 \end{bmatrix}$$

$$x = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\nabla^2 f(x) = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \Rightarrow \text{sign indef}$$

$\Rightarrow$ $x = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ is saddle

$$x = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}$$

$$\nabla^2 f(x) = \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix} > 0$$

$$\det(H_1) = 3$$

$$\det(H_2) = 9$$

$\Rightarrow$ $x = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}$ is a local min

$x = \begin{pmatrix} -1/2 \\ -1/2 \end{pmatrix}$ $\quad D^2 f(x) = \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix} > 0$

$\Rightarrow$ $x = \begin{pmatrix} -1/2 \\ -1/2 \end{pmatrix}$ is a local min

3.) $f(x_1, x_2, x_3) = x_1x_2 + x_2x_3 + x_3x_1 - x_1^3 - x_2^3 - x_3^3$

$$\nabla f = \begin{bmatrix} x_2 + x_3 - 3x_1^2 \\ x_1 + x_3 - 3x_2^2 \\ x_2 + x_3 - 3x_3^2 \end{bmatrix}$$

$\nabla f(x) = 0 \Rightarrow$

$$x_2 + x_3 - 3x_1^2 = 0$$
$$x_1 + x_3 - 3x_2^2 = 0$$
$$x_2 + x_3 - 3x_3^2 = 0$$

$$x_2 - x_1 - 3x_1^2 - 3x_2^2 = 0$$
$$(x_2 - x_1) - 3(x_1 - x_2)(x_1 + x_2) = 0$$
$$(x_2 - x_1)\left[1 + 3(x_1 + x_2)\right] = 0$$

If $x_1 = x_2$,

then $x_1 = x_2 = x_3$ by symmetry $\Rightarrow$

$$x + x - 3x^2 = 0$$
$$-3x^2 + 2x = 0$$
$$x(-3x + 2) = 0$$

$$x_1^* = x_2^* = x_3^* = 0 \quad \text{or} \quad 2/3$$

If $1 + 3(x_1 + x_2) = 0$

$$x_1 + x_2 = -1/3$$
$$-1/3 - 3x_3^2 = 0$$
$$-3x_3^2 = 1/3$$

$\Rightarrow$ No real solutions

SOC

$$\nabla^2 f(x) = \begin{bmatrix} -6x_1 & 1 & 1 \\ 1 & -6x_2 & 1 \\ 1 & 1 & -6x_3 \end{bmatrix}$$

Case $x = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

$$\nabla^2 f(x) = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$\det H_1 = 0$

$\det H_2 = -1$ $\Rightarrow$ sign indef

$\det H_3 = -1 + 1 = 0$

Therefore $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ is a saddle

Case $x^* = \begin{pmatrix} 2/3 \\ 2/3 \\ 2/3 \end{pmatrix}$

$$\nabla^2 f(x^*) = \begin{bmatrix} -4 & 1 & 1 \\ 1 & -4 & 1 \\ 1 & 1 & -4 \end{bmatrix}$$

$\det H_1 = -4$

$\det H_2 = 15$

$\det H_3 = -4(16-1) - 1(-4-1) + 1(16-1) = -40$

$\Rightarrow \nabla^2 f(x^*) \leq 0 \Rightarrow$ x^* is a local max

4.) n factories

Factory i @ $(x_i, y_i) \in \mathbb{R}^2$

Warehouse @ (x, y)

$$\min \sum_{i=1}^{n} (x-x_i)^2 + (y-y_i)^2$$

$$\frac{\partial f}{\partial x} = \sum_{i=1}^{n} 2(x-x_i) \qquad \frac{\partial f}{\partial x} = 0 = 2\left[\sum x - \sum x_i\right] \rightarrow nx = \sum x_i$$

$$x = \frac{1}{n}\sum x_i$$

$$\frac{\partial f}{\partial x} = \sum_{i=1}^{n} 2(y-y_i) \qquad \frac{\partial f}{\partial y} = 0 \rightarrow y = \frac{1}{n}\sum y_i$$

$$\nabla^2 f(x) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} > 0 \Rightarrow$$ all crit pts are local mins

$\therefore$ The optimal soln is $\begin{pmatrix} x^* \\ y^* \end{pmatrix} = \frac{1}{n}\begin{pmatrix} \sum x_i \\ \sum y_i \end{pmatrix}$

5.) $f(a,b) = \sum_{i=1}^{5} (ax_i^2 + by_i^2 - 1)^2$

$$\frac{\partial f}{\partial a} = \sum_{i=1}^{5} 2(ax_i^2 + by_i^2 - 1)(x_i^2)$$

$$\frac{\partial f}{\partial a} = 0 \rightarrow 2\sum_{i=1}^{5} ax_i^4 + by_i^2 x_i^2 - x_i^2 = 0$$

$$= a\sum_{i=1}^{5} x_i^4 + b\sum_{i=1}^{5} y_i^2 x_i^2 = \sum_{i=1}^{5} x_i^2$$

$$\frac{\partial f}{\partial b} = \sum_{i=1}^{5} 2(ax_i^2 + by_i^2 - 1)(y_i^2)$$

$$\frac{\partial f}{\partial b} = 0 \rightarrow a\sum x_i^2 y_i^2 + b\sum y_i^4 = \sum y_i^2$$

$\sum_{i=1}^{5} x_i^4 = 10272$ $\quad$ $\sum_{i=1}^{5} x_i^2 = 120$

$\sum_{i=1}^{5} y_i^4 = 10881$ $\quad$ $\sum_{i=1}^{5} y_i^2 = 141$

$\sum_{i=1}^{5} x_i^2 y_i^2 = 10400$

$$10272a + 10400b = 120$$
$$10400a + 10881b = 141$$

$a \approx -0.0445$, $b \approx 0.0555$

We can notice that error function is convex, because it is the sum of quadratic functions.

$$\nabla^2 f(x) = \begin{bmatrix} 2\sum x_i^4 & 2\sum x_i^2 y_i^2 \\ 2\sum x_i^2 y_i^2 & 2\sum y_i^4 \end{bmatrix} > 0$$

Since the $\det(H_1)$ and $\det(H_2)$ is positive, $\nabla^2 f(x) > 0 \ \forall x$

$\therefore \quad a^* = -0.0455, \quad b^* = 0.0555$